

The International Relations Theory of the Standard Nuclear oliGARCHy

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Abstract

This paper develops an international-relations theory of the Standard Nuclear oliGARCHy (SNoG) by studying a two-economy world in which an incumbent SNoG coexists with a second economy attempting to replicate its institutional, economic, financial, political, strategic, and deterrence architecture. The central result is an Anti-Replication Theorem. Exact replication is not generically an equilibrium outcome when the incumbent possesses previously derived knock-out options and nuclear knock-out options. The ordinary knock-out option makes the object of replication endogenous: the incumbent can change policies, institutions, alliances, financial arrangements, or strategic posture in response to convergence. The nuclear knock-out option adds a catastrophic-risk state whose expected cost may rise with strategic similarity. Consequently, the marginal private return to replication can become negative before complete convergence.

The paper distinguishes technological replication from strategic replication, derives a First-Mover Non-Isomorphism Principle, formulates replication as an optimal-stopping problem and stochastic differential game, and establishes conditions under which the subgame-perfect equilibrium exhibits strategic differentiation rather than institutional cloning. The framework is connected to economics, finance, real-options theory, game theory, political science, international relations, diplomacy, welfare economics, warfare economics, physics, chemistry, biology, psychology, psychiatry, philosophy, statistics, computer science, data science, and artificial intelligence. A computational architecture and empirical research program are provided. The central international-relations implication is that the first SNoG changes the state space in which every subsequent imitation attempt occurs. Thus another economy may imitate observable components of the SNoG, but cannot reproduce the complete strategic object under identical international conditions.

The paper ends with “The End”

Keywords: Standard Nuclear oliGARCHy; SNoG; international relations; deterrence; knock-out option; nuclear knock-out option; strategic replication; first-mover advantage; differential games; real options; geopolitical competition; strategic differentiation; institutional evolution.

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1 Introduction

The Standard Nuclear oliGARCHy (SNoG) has previously been studied as an integrated economic, financial, political, institutional, strategic, and nuclear system. Earlier papers developed its strategic equilibrium, political institutions, knock-out-option logic, nuclear knock-out-option logic, and nuclear diplomacy. The present paper asks a different question:

What happens to the international system if a second economy attempts to replicate the SNoG as a whole?

At first sight, the problem appears to be one of technological diffusion. Suppose economy A has successfully constructed a SNoG architecture and economy B observes its institutions. If institutions are reproducible, knowledge is transferable, capital can be accumulated, and administrative technologies can be learned, one might conjecture that

$$\mathcal{S}_B(t) \longrightarrow \mathcal{S}_A.$$

This conjecture ignores strategic interaction.

The SNoG is not a passive technology. It is a strategic system embedded in an international environment. The incumbent reacts to imitation. Consequently, the object being copied changes as copying proceeds.

This observation leads to the fundamental distinction of this paper:

$\text{technological replicability} \neq \text{strategic replicability}.$

A second economy may reproduce observable institutions without reproducing the geopolitical state in which those institutions operate.

The existence of the first SNoG therefore changes the feasibility set of the second.

The paper establishes five principal results.

First, replication generates an endogenous strategic externality:

$$\mu \uparrow \implies \text{strategic similarity} \uparrow \implies \text{incumbent response} \uparrow.$$

Second, ordinary knock-out options make the incumbent a moving target.

Third, nuclear knock-out options generate a state-dependent catastrophic-risk premium that can make the marginal value of further imitation negative.

Fourth, backward induction removes exact replication from the equilibrium path under suitable credibility and payoff conditions.

Fifth, the international equilibrium is generally characterized not by institutional convergence but by strategic differentiation.

The resulting international system resembles an evolutionary ecology in which competition for the same strategic niche creates incentives for differentiation.

2 Relationship to the Previous SNoG Results

The present paper deliberately does not reconstruct the entire SNoG. Instead, it imports the relevant results from the earlier papers.

Let

$$\mathcal{S}_A = (E_A, F_A, M_A, I_A, P_A, D_A, N_A, \Xi_A),$$

where:

- E_A denotes the real economy;
- F_A denotes the financial architecture;
- M_A denotes the monetary architecture;
- I_A denotes institutional organization;
- P_A denotes political organization;
- D_A denotes conventional defensive capacity;
- N_A denotes nuclear-deterrence capacity;
- Ξ_A denotes the remaining strategic state variables.

The earlier strategic-equilibrium analysis determines how these components interact internally. The political-institutions analysis determines the institutional mechanisms through which collective decisions are made. The nuclear-diplomacy analysis determines how nuclear capability enters bargaining, signalling, deterrence, escalation control, and international negotiation.

The earlier knock-out-option papers provide two objects that are taken as primitive here:

$$K_A$$

and

$$K_A^N,$$

where K_A is the ordinary strategic knock-out option and K_A^N is the nuclear knock-out option. This paper asks what these inherited options imply for international institutional replication.

3 The Two-Economy International System

Consider two strategically independent economies:

$$A = \text{incumbent SNoG},$$

and

$$B = \text{prospective replicator}.$$

Economy A begins with architecture $\mathcal{S}_A$.

Economy B has architecture

$$\mathcal{S}_B(t).$$

Define a replication index

$$\mu(t) \in [0, 1].$$

The endpoints satisfy

$$\mu(t) = 0$$

for complete strategic dissimilarity and

$$\mu(t) = 1$$

for exact replication of the observable architecture.

More generally, let

$$\mu(t) = \sum_{j=1}^J w_j \mu_j(t),$$

where

$$w_j \geq 0, \quad \sum_{j=1}^J w_j = 1.$$

The components may include economic, financial, institutional, political, technological, diplomatic, and strategic similarity.

For example,

$$\mu = w_E \mu_E + w_F \mu_F + w_M \mu_M + w_I \mu_I + w_P \mu_P + w_D \mu_D + w_N \mu_N.$$

The weighting system need not be linear empirically. A generalized aggregator is

$$\mu = \Phi(\mu_E, \mu_F, \mu_M, \mu_I, \mu_P, \mu_D, \mu_N),$$

with

$$\frac{\partial \Phi}{\partial \mu_j} \geq 0.$$

4 The Naive Replication Model

Suppose initially that A is passive.

Then replication may follow

$$d\mu_t = v_B(\mu_t, u_t) dt + \sigma_\mu(\mu_t) dW_t,$$

where u_t denotes replication effort and W_t is a Wiener process.

In a deterministic benchmark,

$$\dot{\mu} = v_B(\mu, u).$$

If

$$v_B(\mu, u) > 0$$

for all $\mu < 1$, then

$$\lim_{t \rightarrow \infty} \mu(t) = 1.$$

This is the institutional-diffusion benchmark.
It is strategically incomplete because it assumes

$$\frac{\partial \mathcal{S}_A}{\partial \mathcal{S}_B} = 0.$$

In an international system this assumption is generally false.
Instead,

$$\frac{\partial \mathcal{S}_A}{\partial \mathcal{S}_B} \neq 0.$$

The incumbent observes the rival.
The rival observes the incumbent.
The state therefore evolves recursively.

5 Strategic Reflexivity

Let σ_A and σ_B denote the strategies of the two economies.
The international state is

$$\Gamma_t = \Gamma(\mathcal{S}_A(t), \mathcal{S}_B(t), \sigma_A(t), \sigma_B(t)).$$

The feedback loop is

$$\mathcal{S}_B \rightarrow \mu \rightarrow \sigma_A \rightarrow \Gamma \rightarrow \sigma_B \rightarrow \mathcal{S}_B.$$

Hence the replication problem is reflexive.

Definition 1 (Strategic Reflexivity). *A replication process is strategically reflexive if the act of increasing similarity changes the incumbent's strategy and thereby changes the environment in which the replicator evaluates further similarity.*

Thus

$$\mathcal{S}_B^* = F(\mathcal{S}_A, \sigma_A, \Gamma),$$

rather than

$$\mathcal{S}_B^* = \mathcal{S}_A.$$

6 Ordinary Knock-Out Options

The previously derived ordinary knock-out option is denoted by

$$K_A(\mu).$$

Suppose there exists a threshold

$$\bar{\mu}_K < 1$$

such that the strategic value of exercising or credibly threatening the ordinary knock-out option rises substantially when

$$\mu \geq \bar{\mu}_K.$$

The option need not correspond to physical destruction. In the international relations interpretation it may represent a lawful and non-destructive strategic response such as institutional redesign, financial restructuring, alliance formation, diplomatic repositioning, technological innovation, economic adaptation, or another change in the incumbent's strategic state.

Write

$$\mathcal{S}'_A = \mathcal{S}_A + \Delta_A.$$

If the change is effective, then

$$\mu(\mathcal{S}_B, \mathcal{S}'_A) < \mu(\mathcal{S}_B, \mathcal{S}_A).$$

The target has moved.

Repeated adaptation produces

$$\mathcal{S}_A \rightarrow \mathcal{S}'_A \rightarrow \mathcal{S}''_A \rightarrow \dots$$

Define incumbent adaptation velocity

$$v_A = \left\| \frac{d\mathcal{S}_A}{dt} \right\|,$$

and rival replication velocity

$$v_B = \left\| \frac{d\mathcal{S}_B}{dt} \right\|.$$

A sufficient dynamic condition for persistent non-convergence is

$$v_A \geq v_B$$

inside the strategic knock-out region.

Proposition 1 (Moving-Target Proposition). *Suppose that whenever $\mu \geq \bar{\mu}_K$, the incumbent can select an adaptation Δ_A such that the resulting decline in similarity weakly offsets the rival's subsequent replication gain. Then exact replication cannot be reached in finite time along the induced strategic path.*

Proof. Let the rival's gross similarity gain over a short interval be $d\mu_B^+ \geq 0$. Let incumbent adaptation induce similarity loss $d\mu_A^- \geq 0$. By assumption,

$$d\mu_A^- \geq d\mu_B^+$$

whenever

$$\mu \geq \bar{\mu}_K.$$

Therefore

$$d\mu = d\mu_B^+ - d\mu_A^- \leq 0.$$

Hence μ cannot cross continuously from the knock-out region to $\mu = 1$. Therefore exact replication is not reached along the induced path. $\square$

7 Nuclear Knock-Out Options as Catastrophic-Risk Options

The nuclear knock-out option is treated here as an inherited deterrence object from the previous SNoG work.

No operational theory of nuclear employment is required.

For the present paper, its economic relevance is entirely captured by the existence of a catastrophic strategic state.

Let

$$p_N(\mu)$$

denote the perceived probability of entering that state conditional on strategic similarity. The central assumption is

$$p'_N(\mu) \geq 0$$

over the strategically sensitive region.

Let

$$L_B^N > 0$$

be the expected loss to economy B conditional on the catastrophic state.

Then the expected nuclear knock-out cost is

$$R_B^N(\mu) = p_N(\mu)L_B^N.$$

Its marginal value is

$$\frac{dR_B^N}{d\mu} = p'_N(\mu)L_B^N.$$

Because L_B^N is potentially very large, even a small derivative $p'_N(\mu)$ can materially affect the optimization problem.

This is the first crucial asymmetry:

$$p'_N(\mu) \text{ small} \not\Rightarrow p'_N(\mu)L_B^N \text{ small.}$$

8 The Economics of Replication

Let

$$V_B(\mu)$$

be the gross economic and strategic value of replication.

Let

$$C_B(\mu)$$

be the direct replication cost.

Let

$$R_B^K(\mu)$$

be the expected cost generated by ordinary knock-out responses.

Let

$$R_B^N(\mu)$$

be the expected cost associated with nuclear knock-out exposure.
Then

$$W_B(\mu) = V_B(\mu) - C_B(\mu) - R_B^K(\mu) - R_B^N(\mu).$$

The marginal payoff is

$$W'_B(\mu) = V'_B(\mu) - C'_B(\mu) - (R_B^K)'(\mu) - (R_B^N)'(\mu).$$

Substituting the catastrophic-risk term gives

$$W'_B(\mu) = V'_B(\mu) - C'_B(\mu) - (R_B^K)'(\mu) - p'_N(\mu)L_B^N.$$

Therefore, if

$$C'_B(\mu) + (R_B^K)'(\mu) + p'_N(\mu)L_B^N > V'_B(\mu),$$

then

$$W'_B(\mu) < 0.$$

At such a point, increasing similarity destroys value.

9 The Replication Paradox

Definition 2 (Replication Paradox). *The replication paradox is the condition under which successful replication raises strategic similarity but simultaneously increases countervailing strategic costs sufficiently rapidly that the expected marginal payoff from additional replication becomes negative.*

The mechanism is

$$\mu \uparrow \Rightarrow \text{similarity} \uparrow \Rightarrow \text{strategic salience} \uparrow \Rightarrow R_B^K + R_B^N \uparrow \Rightarrow W'_B(\mu) \downarrow.$$

The rival can therefore become a victim of its own success.

10 The Anti-Replication Theorem

Theorem 1 (SNoG Anti-Replication Theorem). *Let A be an established SNoG and B a strategically independent economy attempting to replicate it. Suppose:*

1. $V_B(1) < \infty$;
2. $C_B(\mu) \geq 0$;
3. $(R_B^K)'(\mu) \geq 0$ in the strategic replication region;
4. $(R_B^N)'(\mu) \geq 0$ in the strategic replication region;
5. there exists $\bar{\mu} < 1$ such that for every $\mu \in (\bar{\mu}, 1]$,

$$C'_B(\mu) + (R_B^K)'(\mu) + (R_B^N)'(\mu) > V'_B(\mu).$$

Then exact replication is not optimal. In particular,

$$\mu_B^* \leq \bar{\mu} < 1.$$

Proof. The rival maximizes

$$W_B(\mu) = V_B(\mu) - C_B(\mu) - R_B^K(\mu) - R_B^N(\mu).$$

Differentiating,

$$W'_B(\mu) = V'_B(\mu) - C'_B(\mu) - (R_B^K)'(\mu) - (R_B^N)'(\mu).$$

By Assumption 5,

$$W'_B(\mu) < 0$$

for all

$$\mu \in (\bar{\mu}, 1].$$

Hence W_B is strictly decreasing throughout the final replication region. No maximizer can therefore occur at $\mu = 1$. Any optimum satisfies

$$\mu_B^* \leq \bar{\mu} < 1.$$

Thus exact replication is not an optimal strategy. $\square$

11 A Stronger Dynamic Result

Let replication evolve according to

$$d\mu_t = f(\mu_t, u_t, a_t) dt + \sigma(\mu_t) dW_t,$$

where u_t is rival replication effort and a_t is incumbent adaptation.

Economy B solves

$$\max_{\{u_t\}} \mathbb{E}_0 \left[\int_0^\infty e^{-rt} (\pi_B(\mu_t, u_t) - R_B^K(\mu_t) - R_B^N(\mu_t)) dt \right].$$

Economy A solves

$$\max_{\{a_t\}} \mathbb{E}_0 \left[\int_0^\infty e^{-rt} \Pi_A(\mu_t, a_t) dt \right].$$

The Hamilton–Jacobi–Bellman equation for B is

$$rJ_B(\mu) = \max_u \left\{ \pi_B(\mu, u) - R_B^K(\mu) - R_B^N(\mu) + J'_B(\mu) f(\mu, u, a^*) + \frac{1}{2} \sigma^2(\mu) J''_B(\mu) \right\}.$$

The corresponding incumbent equation is

$$rJ_A(\mu) = \max_a \left\{ \Pi_A(\mu, a) + J'_A(\mu) f(\mu, u^*, a) + \frac{1}{2} \sigma^2(\mu) J''_A(\mu) \right\}.$$

The equilibrium is a stochastic differential game.

Exact replication requires $\mu = 1$ to be reachable and optimal under the equilibrium controls.

The Anti-Replication Theorem supplies conditions under which it is not.

12 Optimal Stopping

Replication can alternatively be formulated as an optimal-stopping problem.

Let

$$\tau_B = \inf \{t : B \text{ abandons exact replication}\}.$$

Let τ_N denote entry into the catastrophic strategic state.

Then

$$J_B = \sup_{\tau_B} \mathbb{E}_0 \left[\int_0^{\tau_B} e^{-rt} \pi_B(\mu_t) dt - e^{-r\tau_N} L_B^N \mathbf{1}_{\{\tau_N < \tau_B\}} \right].$$

Let $A_B(\mu)$ denote the value of abandoning exact replication and pursuing a differentiated architecture.

Then the stopping boundary μ^* satisfies the value-matching condition

$$J_B(\mu^*) = A_B(\mu^*).$$

Under regularity conditions, smooth pasting gives

$$J'_B(\mu^*) = A'_B(\mu^*).$$

For

$$\mu > \mu^*,$$

continued replication is dominated by differentiation.

Therefore,

$$\boxed{\mu_B^* < 1.}$$

13 Real-Options Interpretation

The international-relations problem has a direct analogue in financial economics.

A standard knock-out derivative ceases to exist after a barrier is reached. The strategic analogue is not literal identity with the financial contract, but the mathematical structure is useful.

Let the replication project have continuation value X_t .

Suppose a strategic barrier H exists.

Then the value of the replication project may be represented schematically as

$$\mathcal{V}_B = \mathbb{E} \left[e^{-rT} \Phi(X_T) \mathbf{1}_{\{\tau_H > T\}} \right],$$

where

$$\tau_H = \inf \{t : X_t \geq H\}.$$

As the process approaches the barrier, the value of successful continuation need not increase monotonically.

The strategic interpretation is therefore:

greater apparent success $\nRightarrow$ greater option value.

The option can become less valuable precisely because the project approaches a strategically sensitive boundary.

14 Credibility and Deterrence

A threat without credibility cannot support equilibrium deterrence.

Let

$$U_A(K \mid \mu)$$

be the incumbent's payoff from exercising a knock-out response and

$$U_A(T \mid \mu)$$

its payoff from tolerating continued replication.

Credibility requires

$$U_A(K \mid \mu) \geq U_A(T \mid \mu).$$

For the nuclear option, write

$$U_A(K^N \mid \mu) = B_A^N(\mu) - C_A^N(\mu) - D_A^N(\mu),$$

where D_A^N captures expected catastrophic losses to the incumbent itself.

A nuclear option therefore cannot be assumed credible merely because the capability exists.

Instead define

$$\Delta_A^N(\mu) = U_A(K^N \mid \mu) - U_A(T \mid \mu).$$

State-contingent credibility exists when

$$\Delta_A^N(\mu) < 0$$

for low similarity but

$$\Delta_A^N(\mu) \geq 0$$

in some sufficiently extreme strategic state.

Thus deterrence may operate through the rival's anticipation of a future change in incentives.

15 Backward Induction

Consider a finite approximation with replication stages

$$0 = \mu_0 < \mu_1 < \cdots < \mu_M = 1.$$

At stage $M - 1$, economy B chooses between:

Complete

and

Differentiate.

If

$$U_B(\text{Differentiate} \mid \mu_{M-1}) > U_B(\text{Complete} \mid \mu_{M-1}),$$

then the final transition is rejected.

Hence

$$\mu_M = 1$$

is not reached.

Backward induction then changes the value of investment at μ_{M-2} , μ_{M-3} , and earlier stages.

Proposition 2 (Backward-Induction Non-Replication). *If the final replication step is strictly dominated by strategic differentiation and all players possess common knowledge of rationality, exact replication is absent from the subgame-perfect equilibrium path.*

Proof. At the final decision node, rationality implies differentiation. Therefore the history terminating in exact replication cannot be generated from that node. Common knowledge implies that earlier players correctly anticipate this decision. Hence no subgame-perfect equilibrium can prescribe exact replication as the terminal continuation of the relevant subgame. $\square$

Thus, in the strict model,

$$\boxed{\mathbb{P}(\mu = 1 \mid \text{SPE}) = 0.}$$

16 Technological Replication Versus Strategic Replication

The strongest version of the argument does not require the claim that institutions are physically impossible to copy.

Suppose instead that

$$\mathcal{S}_B = \mathcal{S}_A.$$

Let

$$\Gamma_A$$

and

$$\Gamma_B$$

denote the international environments occupied by the two systems.

Define the complete strategic object as

$$\tilde{\mathcal{S}}_i = (\mathcal{S}_i, \Gamma_i).$$

Even if

$$\mathcal{S}_A = \mathcal{S}_B,$$

it does not follow that

$$\tilde{\mathcal{S}}_A = \tilde{\mathcal{S}}_B.$$

Indeed, the first system changes the environment encountered by the second.

17 The First-Mover Non-Isomorphism Principle

Suppose the original SNoG emerges in international state

$$\Gamma_0.$$

Its existence transforms the international system according to

$$\Gamma_1 = T(\Gamma_0, \mathcal{S}_A).$$

The original complete state is

$$\tilde{\mathcal{S}}_A = (\mathcal{S}_A, \Gamma_0).$$

A literal institutional copy created afterward would instead be

$$\tilde{\mathcal{S}}_B = (\mathcal{S}_A, \Gamma_1).$$

If

$$\Gamma_1 \neq \Gamma_0,$$

then

$$\tilde{\mathcal{S}}_B \neq \tilde{\mathcal{S}}_A.$$

Theorem 2 (First-Mover Non-Isomorphism Principle). *If the creation of the first SNoG changes the international state, then no subsequent economy can reproduce the complete initial SNoG strategic state merely by reproducing its observable domestic architecture.*

Proof. The first complete strategic object is

$$(\mathcal{S}_A, \Gamma_0).$$

By hypothesis,

$$T(\Gamma_0, \mathcal{S}_A) = \Gamma_1$$

with

$$\Gamma_1 \neq \Gamma_0.$$

A subsequent institutional copy therefore occupies

$$(\mathcal{S}_A, \Gamma_1).$$

Since the second coordinates differ,

$$(\mathcal{S}_A, \Gamma_1) \neq (\mathcal{S}_A, \Gamma_0).$$

Hence the complete strategic objects are non-isomorphic under identity of their environmental state. $\square$

This theorem is conceptually deeper than a simple copying-cost argument.

The first SNoG changes the state space.

18 International Relations Interpretation

The model combines several major traditions in international relations.

18.1 Realism

The realist interpretation emphasizes strategic competition, survival, relative capabilities, uncertainty, and security externalities.

Replication is threatening not merely because B becomes richer, but because its relative strategic position changes:

$$\frac{d}{d\mu} \left(\frac{\mathcal{P}_B}{\mathcal{P}_A} \right) > 0,$$

where $\mathcal{P}_i$ denotes composite power.

18.2 Security Dilemma

Suppose B views replication as defensive:

$$\text{Intent}_B = \text{security}.$$

Economy A may nevertheless observe

$$\text{Capability}_B \uparrow.$$

Thus

$$\text{defensive intention} + \text{offensive capability ambiguity} \rightarrow \text{security dilemma}.$$

The result can be inefficient strategic competition even when neither side initially seeks confrontation.

18.3 Liberal Institutionalism

Institutions can reduce the sensitivity of knock-out risk to similarity.

Let q denote institutionalized cooperation.

Then write

$$p_N = p_N(\mu, q),$$

with

$$\frac{\partial p_N}{\partial q} < 0.$$

Arms-control arrangements, verification, communication, transparency, commercial interdependence, and diplomatic institutions can therefore shift the stopping boundary.

18.4 Constructivism

Strategic similarity is partly socially interpreted.

Let

$$\hat{\mu} = g(\mu, \mathcal{N}, \mathcal{I}),$$

where $\mathcal{N}$ represents narratives and $\mathcal{I}$ represents identity structures.

Then policy responds to perceived similarity,

$$p_N = p_N(\hat{\mu}),$$

rather than purely material similarity.

Thus discourse and identity can affect equilibrium outcomes.

19 Diplomacy as a Risk-Transformation Technology

Diplomacy is not merely communication in this model. It changes the mapping from material similarity into strategic risk.

Let diplomatic effort be d .

Then

$$p_N = p_N(\mu, d).$$

Assume

$$\frac{\partial p_N}{\partial d} < 0.$$

The rival's payoff becomes

$$W_B(\mu, d) = V_B(\mu) - C_B(\mu) - R_B^K(\mu, d) - p_N(\mu, d)L_B^N - C_d(d).$$

The optimal diplomatic effort satisfies

$$-\frac{\partial R_B^K}{\partial d} - \frac{\partial p_N}{\partial d} L_B^N = C'_d(d)$$

at an interior solution.

Diplomacy therefore has a shadow price equal to the marginal reduction in expected strategic loss.

This connects diplomacy directly to welfare economics and financial risk management.

20 Bargaining Theory

Let the bargaining surplus from peaceful differentiation be

$$S = V_A^D + V_B^D - V_A^C - V_B^C,$$

where D denotes negotiated differentiation and C denotes continued strategic competition. If

$$S > 0,$$

there exists a bargaining range.

Let $\theta \in [0, 1]$ denote bargaining power of A .

A Nash bargaining allocation can be represented as

$$x_A = \theta S,$$

$$x_B = (1 - \theta)S.$$

Thus knock-out-option competition need not culminate in confrontation.

Indeed, if the options are credible, they may increase incentives to negotiate before the costly region is reached.

The paradox is therefore:

credible destructive possibility $\rightarrow$ greater value of peaceful bargaining.

21 Welfare Economics

Private incentives and global welfare need not coincide.

Let

$$W^G = W_A + W_B - \mathcal{X},$$

where $\mathcal{X}$ denotes international externalities.

Suppose competitive replication raises both economies' security expenditures.

Then

$$\frac{\partial W_A}{\partial \mu}$$

and

$$\frac{\partial W_B}{\partial \mu}$$

may fail to internalize

$$\frac{\partial \mathcal{X}}{\partial \mu}.$$

The global optimum solves

$$\frac{dW^G}{d\mu} = 0.$$

Thus

$$V'_B(\mu) - C'_B(\mu) - (R_B^K)'(\mu) - (R_B^N)'(\mu) - \mathcal{X}'(\mu) = 0.$$

The socially optimal similarity level may therefore be lower than the uncoordinated private level.

22 Warfare Economics

Let total resources be

$$Y_i = C_i + I_i + G_i^S,$$

where C_i is civilian consumption, I_i investment, and G_i^S strategic expenditure.

Replication-induced competition can increase

$$G_A^S \quad \text{and} \quad G_B^S.$$

Hence

$$\frac{\partial C_i}{\partial G_i^S} < 0$$

holding output fixed.

The opportunity cost of strategic competition is therefore

$$OC_i = \Delta C_i + \Delta I_i.$$

The model predicts that strategic differentiation can dominate exact replication even before catastrophic risk is considered because duplication can induce persistent resource diversion.

23 Finance and Asset Pricing

Strategic replication affects financial prices through risk premia.

Let an asset return satisfy

$$R_{t+1} = R_f + \beta_M \lambda_M + \beta_G \lambda_G + \epsilon_{t+1},$$

where λ_G is a geopolitical-risk premium.

Suppose

$$\lambda_G = \lambda_G(\mu)$$

with

$$\lambda'_G(\mu) > 0$$

inside the sensitive region.

Then the cost of capital for economy B can rise with replication:

$$r_B(\mu) = r_f + \lambda_G(\mu) + \lambda_B.$$

Investment value becomes

$$NPV_B(\mu) = \sum_{t=1}^T \frac{\mathbb{E}[CF_t(\mu)]}{(1 + r_B(\mu))^t} - I_0(\mu).$$

Thus geopolitical competition creates a financial feedback loop:

$$\mu \uparrow \rightarrow \lambda_G \uparrow \rightarrow r_B \uparrow \rightarrow NPV_B \downarrow.$$

This mechanism strengthens the Anti-Replication Theorem.

24 Physics Analogy: Repulsive Strategic Potential

The model has a useful mathematical analogy to a potential field.

Let $U(\mu)$ denote strategic potential.

Define the force

$$F(\mu) = -\frac{dU}{d\mu}.$$

Suppose the incumbent's deterrence structure creates a steep potential near $\mu = 1$:

$$\lim_{\mu \rightarrow 1^-} U(\mu) = +\infty.$$

For example,

$$U(\mu) = \frac{\kappa}{(1 - \mu)^\alpha}, \quad \kappa > 0, \quad \alpha > 0.$$

Then

$$F(\mu) = -\frac{\kappa\alpha}{(1 - \mu)^{\alpha+1}}.$$

The magnitude of the repulsive force rises sharply near exact replication.

This is an analogy, not a physical claim. It illustrates mathematically how the final increments of strategic convergence may become increasingly costly.

25 Chemistry Analogy: Activation Barriers

Consider a reaction-coordinate analogy.

Let μ represent progress toward a new strategic configuration.

Suppose the system must cross an activation barrier

$$E_a(\mu).$$

The probability of transition can be represented schematically by

$$P_{\text{transition}} \propto \exp\left(-\frac{E_a}{\Theta}\right),$$

where Θ is an effective uncertainty or fluctuation parameter.

If strategic countermeasures increase the barrier,

$$\frac{dE_a}{d\mu} > 0,$$

then convergence becomes progressively less probable.

The analogy emphasizes that the final state can be technologically conceivable while dynamically difficult to reach.

26 Biology and Evolutionary Ecology

The international system can be interpreted as an evolutionary game.

Let x_j denote the share of states using strategic architecture j .

The replicator equation is

$$\dot{x}_j = x_j (f_j - \bar{f}),$$

where

$$\bar{f} = \sum_k x_k f_k.$$

If the fitness of copying the incumbent falls as the number of similar systems increases, then

$$\frac{\partial f_{\text{copy}}}{\partial x_{\text{copy}}} < 0.$$

This is negative frequency dependence.

It supports strategic diversity.

The international equilibrium may therefore exhibit

$$\mathcal{S}_1^* \neq \mathcal{S}_2^* \neq \cdots \neq \mathcal{S}_n^*.$$

The analogue is ecological niche differentiation.

27 Psychology and Psychiatry: Perception Without Pathologization

Strategic decision makers do not observe objective probabilities perfectly.

Let perceived nuclear risk be

$$\hat{p}_N = h(p_N, \eta, \beta, \chi),$$

where η represents information quality, β cognitive bias, and χ stress-dependent distortion. Expected utility becomes

$$\widehat{W}_B = V_B - C_B - R_B^K - \widehat{p}_N L_B^N.$$

Probability weighting can be represented by

$$w(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{1/\gamma}}.$$

Then

$$R_B^N = w(p_N) L_B^N.$$

Small catastrophic probabilities may therefore receive disproportionate decision weight.

The psychiatric relevance is institutional rather than diagnostic: high-stress decision environments may affect cognition, information processing, group dynamics, and judgement. The model makes no inference about the mental health of any particular leader.

28 Philosophy: Identity, Copying, and Strategic Objects

The First-Mover Non-Isomorphism Principle raises a philosophical question: what does it mean to copy a political-economic system?

If identity is purely structural, then

$$\mathcal{S}_A = \mathcal{S}_B$$

might appear sufficient.

If identity is relational, however, the relevant object is

$$\tilde{\mathcal{S}} = (\text{internal structure, external relations}).$$

Then equality requires both

$$\mathcal{S}_A = \mathcal{S}_B$$

and

$$\Gamma_A = \Gamma_B.$$

But the first mover changes Γ .

Therefore structural identity does not imply relational identity.

This is the philosophical foundation of strategic non-replicability.

29 Political Science and Institutional Endogeneity

Domestic institutions respond to external pressure.

Let political architecture satisfy

$$P_i^* = P_i(Y_i, \Gamma_i, T_i, L_i, \Omega_i),$$

where T_i denotes external threat, L_i domestic legitimacy, and Ω_i administrative capacity.

If replication changes external threat,

$$\frac{\partial T_B}{\partial \mu} > 0,$$

then

$$\frac{dP_B^*}{d\mu} = \frac{\partial P_B}{\partial T_B} \frac{\partial T_B}{\partial \mu} + \dots$$

Thus even a government attempting to reproduce P_A may endogenously evolve away from it because its external environment differs.

30 Geopolitics and Alliance Formation

Introduce third states

$$C_1, \dots, C_m.$$

Let the international network be

$$G = (V, E),$$

with

$$V = \{A, B, C_1, \dots, C_m\}.$$

The edge weight

$$g_{ij}$$

measures strategic cooperation.

As B becomes more similar to A , third states may rebalance.

Thus

$$g_{ij} = g_{ij}(\mu).$$

The alliance network itself becomes endogenous.

The second economy therefore cannot replicate the first economy's alliance position because its replication attempt changes the network.

31 Network-Theoretic Non-Replication

Let A_G denote the adjacency matrix of the international system.

The complete state is now

$$\tilde{\mathcal{S}}_i = (\mathcal{S}_i, A_G).$$

If creation of the first SNoG changes at least one edge,

$$A_G^{(1)} \neq A_G^{(0)}.$$

A second SNoG therefore emerges under a different graph.

Graph centrality measures can differ even if domestic institutions are identical.

For example, eigenvector centrality satisfies

$$A_G x = \lambda x.$$

Changing A_G changes x .

Hence identical domestic structures can occupy non-identical geopolitical positions.

32 Information Theory

Replication requires information.

Let X_A denote the true SNoG architecture and Y_B the rival's observation.

The information available for replication is bounded by mutual information

$$I(X_A; Y_B) = H(X_A) - H(X_A | Y_B).$$

If some institutional knowledge is tacit, endogenous, or unobservable, then

$$H(X_A | Y_B) > 0.$$

Consequently,

$$I(X_A; Y_B) < H(X_A).$$

Even without strategic opposition, imperfect information prevents perfect reconstruction.

Strategic adaptation increases the difficulty because the target distribution changes over time.

33 Control Theory

Let the state vector be

$$x_t = \begin{pmatrix} \mu_t \\ q_t \\ p_t \\ g_t \end{pmatrix},$$

where q_t is diplomatic cooperation, p_t strategic risk, and g_t geopolitical tension.

A linearized system is

$$\dot{x}_t = Ax_t + B_A u_{A,t} + B_B u_{B,t}.$$

Economy B may minimize

$$J_B = \int_0^\infty e^{-rt} [(x_t - x_B^*)' Q_B (x_t - x_B^*) + u_{B,t}' R_B u_{B,t}] dt.$$

If the desired target x_B^* includes exact replication but the incumbent's feedback control changes the transition matrix, the closed-loop system becomes

$$\dot{x} = (A - B_A K_A - B_B K_B)x.$$

The reachable set can therefore exclude the exact replication state.

34 Statistics and Econometrics

The theory generates empirical hypotheses.

Let country i at time t have similarity index μ_{it} .

A panel specification is

$$Risk_{it} = \alpha_i + \tau_t + \beta_1 \mu_{it} + \beta_2 \mu_{it}^2 + \gamma' X_{it} + \epsilon_{it}.$$

The theory predicts, in the strategically sensitive region,

$$\beta_1 > 0$$

or, more generally,

$$\frac{\partial Risk_{it}}{\partial \mu_{it}} > 0.$$

A differentiation equation can be estimated as

$$\Delta D_{it+1} = \alpha + \beta Risk_{it} + \gamma \mu_{it} + \delta X_{it} + \epsilon_{it+1},$$

where D measures institutional differentiation.

The prediction is

$$\beta > 0.$$

Strategic risk should predict subsequent differentiation.

35 Survival Analysis

Define the “failure” event as abandonment of exact replication.

Let

$$T_i$$

be the duration of replication.

The hazard function is

$$h(t | X) = h_0(t) \exp(X' \beta).$$

Include similarity as a covariate:

$$h(t) = h_0(t) \exp(\beta_\mu \mu_t + \beta_R R_t + \beta_D D_t).$$

The theory predicts

$$\beta_\mu > 0$$

once observations enter the strategic sensitivity region.

36 Bayesian Learning

Economy B does not know the true response function of A .

Let θ_A denote the incumbent’s strategic type.

The rival begins with prior

$$p(\theta_A).$$

After observing response y_t ,

$$p(\theta_A | y_{1:t}) \propto p(y_t | \theta_A) p(\theta_A | y_{1:t-1}).$$

The perceived knock-out probability becomes

$$\hat{p}_N(\mu_t) = \int p_N(\mu_t, \theta_A) p(\theta_A | y_{1:t}) d\theta_A.$$

Thus strategic interaction is simultaneously a learning process.

37 Machine Learning Interpretation

The replication problem can be viewed as non-stationary learning.

Economy B attempts to learn a policy

$$\pi_B(a \mid s)$$

from observations of A .

But A adapts.

Hence the data-generating process satisfies

$$P_{t+1}(s' \mid s, a) \neq P_t(s' \mid s, a).$$

The environment is non-stationary.

A model trained on historical SNoG behavior therefore suffers concept drift.

Let prediction error be

$$L_t = \left\| \hat{\mathcal{S}}_A(t) - \mathcal{S}_A(t) \right\|^2.$$

If the incumbent adapts faster than the learner updates, then

$$L_t \rightarrow 0.$$

This is the computational analogue of the moving-target proposition.

38 Reinforcement Learning Formulation

Define a Markov game with state

$$s_t = (\mu_t, \Gamma_t, q_t, R_t).$$

Economy B chooses

$$a_t^B \in \{\text{replicate, differentiate, negotiate, pause}\}.$$

Economy A chooses

$$a_t^A \in \{\text{tolerate, adapt, negotiate, signal}\}.$$

The value function is

$$Q_B(s, a_B, a_A) = r_B(s, a_B, a_A) + \gamma \mathbb{E} [V_B(s') \mid s, a_B, a_A].$$

The equilibrium policy is

$$\pi_B^* = \arg \max_{\pi_B} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t^B \right].$$

The Anti-Replication Theorem predicts that sufficiently close to the strategic boundary,

$$\pi_B^*(\text{differentiate} \mid s) > \pi_B^*(\text{replicate} \mid s).$$

39 A Safe Computational Algorithm

The theory can be simulated without modelling operational weapon use. The nuclear component enters only as an abstract catastrophic-risk cost.

Algorithm: SNoG Strategic Replication Game

Input:

Initial similarity μ
Replication cost $C(\mu)$
Replication benefit $V(\mu)$
Ordinary strategic-risk function $RK(\mu)$
Catastrophic-risk function $RN(\mu)$
Incumbent adaptation function $A(\mu)$
Discount factor β
Maximum iterations T

For $t = 1, \dots, T$:

1. Observe current similarity μ .
2. Compute:
 $\text{marginal_benefit} = dV/d\mu$
 $\text{marginal_cost} =$
 $dC/d\mu + dRK/d\mu + dRN/d\mu$
3. If $\text{marginal_cost} > \text{marginal_benefit}$:
 $\text{action_B} = \text{DIFFERENTIATE}$
Else:
 $\text{action_B} = \text{REPLICATE}$
4. If $\text{action_B} == \text{REPLICATE}$:
 incumbent selects lawful strategic adaptation
 update target architecture
 update similarity μ
5. If $\text{action_B} == \text{DIFFERENTIATE}$:
 move B toward its own optimal architecture
6. Update diplomacy, risk beliefs, and international state.
7. Stop if policy converges.

Output:

Equilibrium similarity μ^*
Equilibrium differentiation
Welfare
Strategic-risk path

40 Data-Science Architecture

An empirical implementation requires four broad data layers.

Table 1: Empirical Data Architecture

Layer	Variables	Purpose
Economic	GDP, trade, productivity, investment, fiscal capacity	Measure economic similarity and opportunity costs
Financial	interest rates, sovereign spreads, market volatility, capital flows	Estimate geopolitical risk premia
Institutional	governance structure, legal institutions, administrative capacity	Construct institutional similarity
International	alliances, treaties, diplomatic events, sanctions, disputes	Measure strategic response and differentiation

A composite similarity index may be constructed using standardized variables. For variable j ,

$$z_{ijt} = \frac{x_{ijt} - \bar{x}_j}{s_j}.$$

Distance from the incumbent is

$$d_{Bt} = \sqrt{\sum_j w_j (z_{Bjt} - z_{Ajt})^2}.$$

A bounded similarity measure is

$$\mu_{Bt} = \exp(-d_{Bt}).$$

Then

$$0 < \mu_{Bt} \leq 1.$$

41 Causal Inference

Simple correlations between similarity and strategic tension are insufficient.

The principal endogeneity problem is

$$\mu_{it} \leftrightarrow Risk_{it}.$$

Strategic risk may cause institutional adaptation, while institutional adaptation may cause strategic risk.

Possible empirical strategies include:

1. event studies around major institutional changes;
2. synthetic-control designs;
3. difference-in-differences where credible;
4. instrumental-variable strategies based on plausibly exogenous diffusion shocks;
5. panel vector autoregressions;
6. local projections;

7. Bayesian structural models.

The empirical objective is not to prove the theoretical mechanism from a single coefficient but to test its distinct observable implications.

42 AI-Assisted Strategic Forecasting

Let the state vector be x_t .

A predictive model estimates

$$\hat{x}_{t+h} = f_{\theta}(x_t, x_{t-1}, \dots).$$

The model should output distributions rather than point predictions:

$$p_{\theta}(x_{t+h} \mid x_{\leq t}).$$

Catastrophic-risk applications require calibration.

For predicted probability $\hat{p}$, calibration requires approximately

$$\mathbb{P}(Y = 1 \mid \hat{p} = p) = p.$$

Because geopolitical regimes change, the model should be subjected to out-of-distribution testing.

The appropriate AI role is decision support, not autonomous coercive decision-making.

43 International-System Equilibrium

Now introduce n potential replicators.

Each economy i solves

$$\max_{\mu_i} [V_i(\mu_i) - C_i(\mu_i) - R_i^K(\mu_i) - R_i^N(\mu_i)].$$

Suppose the Anti-Replication condition holds for each.

Then

$$\mu_i^* < 1.$$

Hence exact convergence on the incumbent architecture does not occur.

Instead,

$$\boxed{\mathcal{S}_1^* \neq \mathcal{S}_2^* \neq \dots \neq \mathcal{S}_n^*}$$

This is the Strategic Differentiation Equilibrium.

Definition 3 (Strategic Differentiation Equilibrium). *A strategic differentiation equilibrium is an international equilibrium in which each major economy chooses an architecture that remains distinguishable from the incumbent because the expected marginal cost of exact convergence exceeds its marginal benefit.*

44 Institutional Biodiversity

Let diversity be measured by

$$\mathcal{D} = \sum_{i < j} \omega_{ij} d(\mathcal{S}_i, \mathcal{S}_j).$$

The model implies that knock-out-option competition can generate

$$\mathcal{D} > 0.$$

The international system therefore contains endogenous institutional biodiversity. This is not necessarily globally inefficient.

Diversity may provide:

- experimentation;
- resilience;
- reduced common-mode failure;
- alternative governance technologies;
- reduced direct strategic substitutability.

45 Systemic Risk

Exact institutional replication can create correlated failure.

Let loss in economy i be L_i .

Systemic loss is

$$L^{sys} = \sum_i L_i.$$

Then

$$\text{Var}(L^{sys}) = \sum_i \text{Var}(L_i) + 2 \sum_{i < j} \text{Cov}(L_i, L_j).$$

If institutional cloning increases covariance,

$$\frac{\partial \text{Cov}(L_i, L_j)}{\partial \mu_{ij}} > 0,$$

then differentiation may reduce systemic risk.

Thus strategic differentiation may possess a global diversification benefit.

46 A General Stability Condition

Let the dynamic similarity equation be

$$\dot{\mu} = F(\mu).$$

An interior equilibrium μ^* satisfies

$$F(\mu^*) = 0.$$

Local stability requires

$$F'(\mu^*) < 0.$$

Suppose

$$F(\mu) = v_B(\mu) - v_A(\mu) - \rho(\mu),$$

where $\rho(\mu)$ captures endogenous strategic deterrence.

Then

$$F'(\mu) = v'_B(\mu) - v'_A(\mu) - \rho'(\mu).$$

A sufficient condition for local stability is

$$v'_A(\mu^*) + \rho'(\mu^*) > v'_B(\mu^*).$$

The equilibrium similarity is therefore stabilized below unity when incumbent adaptation and strategic-risk feedback rise sufficiently quickly.

47 Comparative Statics

Let the equilibrium satisfy

$$G(\mu^*, \theta) = 0.$$

By the implicit-function theorem,

$$\frac{d\mu^*}{d\theta} = -\frac{G_\theta}{G_\mu}.$$

Several comparative-static predictions follow.

If diplomacy lowers strategic risk, then typically

$$\frac{d\mu^*}{dd} > 0.$$

If catastrophic loss rises,

$$\frac{d\mu^*}{dL_B^N} < 0.$$

If institutional compatibility raises the benefits of convergence,

$$\frac{d\mu^*}{dV_B'} > 0.$$

If incumbent adaptation becomes faster,

$$\frac{d\mu^*}{dv_A} < 0.$$

Thus peaceful institutions can permit greater benign similarity while strategic rivalry pushes the equilibrium toward differentiation.

48 International Law and Normative Constraints

The feasible strategic set must be restricted by law and normative constraints.

Let

$$\mathcal{A}_A^{legal} \subseteq \mathcal{A}_A$$

denote legally admissible actions.

Then the incumbent solves

$$\max_{a_A \in \mathcal{A}_A^{legal}} U_A(a_A).$$

Likewise for B .

This distinction is important because the theoretical existence of a strategic option does not imply that every conceivable implementation is lawful, credible, desirable, or welfare improving.

The international-relations theory developed here is therefore compatible with a strong emphasis on diplomacy, restraint, verification, communication, and catastrophic-risk reduction.

49 The Role of Nuclear Weapons in the Theory

The nuclear component has a precise and limited analytical role.

It enters as

$$R_B^N(\mu) = p_N(\mu)L_B^N.$$

The theory does not require an operational model of nuclear conflict.

Indeed, because

$$L_B^N$$

and corresponding global losses can be enormous, the principal policy implication is avoidance.

The nuclear knock-out option matters because the *possibility* of a catastrophic state can affect bargaining before that state occurs.

Thus:

option value $\neq$ exercise.

In deterrence theory, an option can affect equilibrium behavior precisely because rational actors seek to avoid reaching the state in which it would become relevant.

50 Diplomatic Escape from the Replication Trap

The Anti-Replication Theorem does not imply inevitable hostility.

There are at least four peaceful equilibrium responses:

1. strategic differentiation;
2. negotiated coexistence;
3. institutional interoperability without identity;
4. cooperative specialization.

Let D_B denote differentiation and Q_{AB} cooperation.
A peaceful equilibrium may therefore satisfy

$$\mu_B^* < 1, \quad Q_{AB}^* > 0.$$

Thus difference and cooperation are compatible.
This is an important normative result.
The theory does not require

$$\text{difference} \Rightarrow \text{hostility}.$$

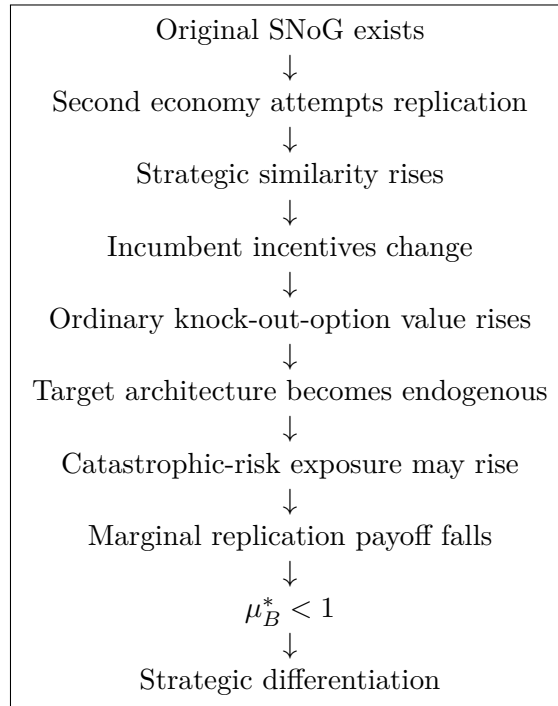
Instead,

$$\text{difference} + \text{credible diplomacy} \Rightarrow \text{stable coexistence}$$

is possible.

51 The Core Causal Chain

The entire theory can be summarized as



52 Vector Graphic I: Strategic Replication Mechanism

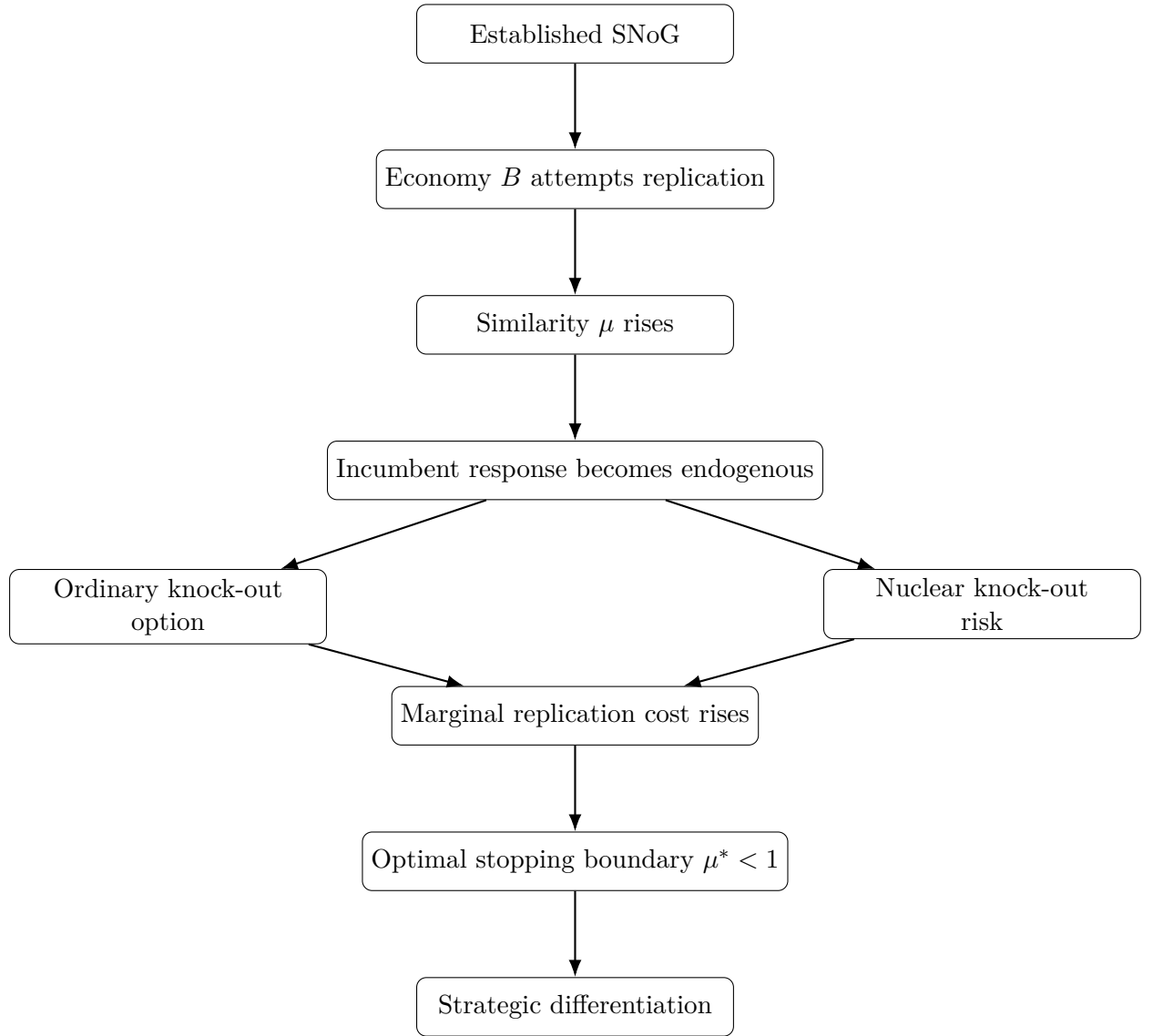


Figure 1: The strategic anti-replication mechanism.

53 Vector Graphic II: The Moving Target

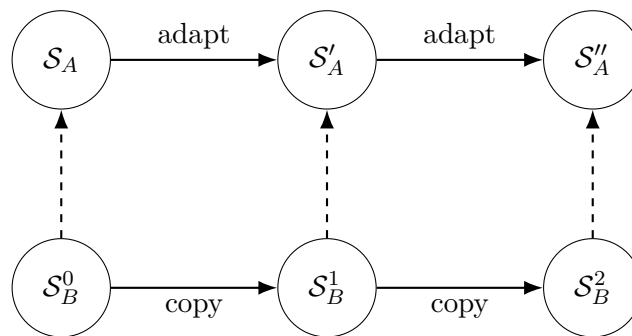


Figure 2: Replication against an endogenously moving strategic target.

54 Vector Graphic III: First-Mover Non-Isomorphism

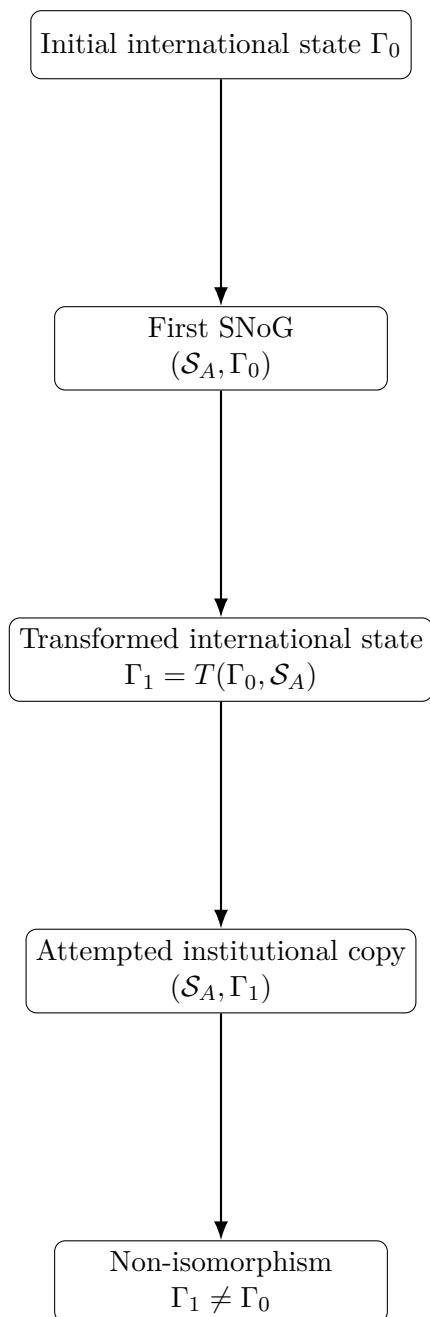


Figure 3: Why identical domestic architecture does not imply identical strategic state.

55 Summary Table of Mechanisms

Table 2: Interdisciplinary Interpretation of Strategic Non-Replication

Field	Mechanism	SNoG Interpretation
Economics	Marginal cost exceeds marginal benefit	Optimal replication stops below unity
Finance	Barrier option and risk premium	Approaching strategic thresholds can destroy project value
Game theory	Backward induction	Final replication step can disappear from SPE
International relations	Security dilemma and deterrence	Similarity changes incumbent incentives
Diplomacy	Risk transformation	Negotiation reduces strategic-risk sensitivity
Political science	Institutional endogeneity	External pressure changes domestic institutions
Physics	Repulsive potential analogy	Resistance rises near the strategic boundary
Chemistry	Activation barrier analogy	Final transition may become dynamically inaccessible
Biology	Negative frequency dependence	Strategic niches encourage differentiation
Psychology	Probability weighting	Small catastrophic probabilities can have large behavioral effects
Philosophy	Relational identity	Structural copying does not reproduce strategic identity
Statistics	Hazard and panel models	Replication hypotheses become empirically testable
Computer science	Moving-target learning	The target changes as the learner approaches it
AI/ML	Non-stationary Markov game	Adaptive policies replace static copying

56 Main Propositions

The theory can now be condensed into a sequence of propositions.

Proposition 3 (Strategic Externality). *If increasing similarity changes the incumbent's expected payoff, then replication generates an international strategic externality.*

Proof. If

$$\frac{\partial U_A}{\partial \mu} \neq 0,$$

then a marginal replication decision by B changes A 's welfare. Since B does not generally internalize the full change in A 's welfare, the decision generates an externality. $\square$

Proposition 4 (Endogenous Target). *If the incumbent's optimal architecture depends on rival similarity, then the replication target is endogenous.*

Proof. Suppose

$$\mathcal{S}_A^* = F_A(\mu).$$

If

$$F'_A(\mu) \neq 0,$$

then changes in replication alter the target architecture. Hence the target is not exogenous. $\square$

Proposition 5 (Catastrophic-Risk Amplification). *If $p'_N(\mu) > 0$ and L_B^N is sufficiently large, catastrophic-risk exposure can dominate the marginal benefit of replication even when $p'_N(\mu)$ is small.*

Proof. The marginal expected catastrophic loss is

$$p'_N(\mu)L_B^N.$$

For any finite positive marginal net benefit $M(\mu)$ and any $p'_N(\mu) > 0$, there exists

$$L_B^N > \frac{M(\mu)}{p'_N(\mu)}$$

such that

$$p'_N(\mu)L_B^N > M(\mu).$$

Therefore catastrophic-risk exposure dominates the marginal benefit. $\square$

Proposition 6 (Strategic Differentiation). *If exact replication is dominated while some differentiated architecture produces positive value, the rival optimally selects a non-identical strategic architecture.*

Proof. Let exact replication yield $W_B(1)$ and some differentiated architecture $\mathcal{S}_B^D$ yield W_B^D . If

$$W_B^D > W_B(1),$$

then exact replication cannot maximize B 's payoff. Therefore an optimum can be selected from the differentiated strategy set. $\square$

57 The Central International Relations Theorem

Theorem 3 (International Relations Theorem of the SNoG). *Suppose:*

1. *an incumbent SNoG changes the international environment by existing;*
2. *a second economy's attempt at replication changes the incumbent's incentives;*
3. *the incumbent possesses ordinary strategic adaptation or knock-out options;*
4. *catastrophic-risk exposure is non-decreasing over a strategically sensitive similarity region;*
5. *the expected marginal strategic cost eventually exceeds the marginal benefit of further replication.*

Then a strategically independent second economy cannot reproduce the complete strategic state of the original SNoG as an equilibrium continuation. The equilibrium response is partial imitation, negotiated coexistence, or strategic differentiation rather than exact strategic replication.

Proof. By Condition 1 and the First-Mover Non-Isomorphism Principle, identical observable institutions do not reproduce the original international state.

By Conditions 2 and 3, the target of replication becomes endogenous.

By Condition 4, increasing similarity weakly increases expected catastrophic-risk exposure in the sensitive region.

By Condition 5, there exists $\bar{\mu} < 1$ such that

$$W'_B(\mu) < 0$$

for

$$\mu > \bar{\mu}.$$

Therefore exact replication at $\mu = 1$ cannot maximize B 's expected payoff.

Since the complete strategic state requires both domestic architectural identity and relational international identity, and neither condition can generically be achieved along the equilibrium path, complete strategic replication fails.

The remaining equilibrium strategies consist of partial imitation, cooperation, negotiated coexistence, or differentiation. $\square$

58 What the Theorem Does Not Claim

Precision is essential.

The theorem does not claim that another society is physically incapable of copying individual institutions.

It does not claim that knowledge cannot diffuse.

It does not claim that economic development by another country is inherently threatening.

It does not claim that catastrophic conflict is inevitable.

It does not claim that possession of an option implies exercise.

Instead, it establishes the narrower and stronger strategic statement:

The existence of the first SNoG changes the optimization problem of every attempted second SNoG.

Therefore,

institutional identity $\nRightarrow$ strategic identity.

59 Policy Implications

The theory generates several policy implications.

First, exact institutional cloning is unnecessary for economic learning. States can adopt useful components without reproducing the complete strategic architecture.

Second, diplomacy should seek to reduce

$$\frac{\partial p_N}{\partial \mu}.$$

Third, transparency can separate benign institutional learning from hostile strategic interpretation.

Fourth, interoperability may dominate identity.

Fifth, differentiated systems can coexist peacefully when bargaining institutions are sufficiently strong.

Sixth, the largest welfare gains arise from keeping catastrophic states off the realized equilibrium path.

Thus the ideal international outcome is not the exercise of knock-out options.

It is an equilibrium in which their existence, together with diplomacy and credible institutions, makes destructive escalation unnecessary.

60 Conclusion

This paper has developed the international-relations theory of the Standard Nuclear oligARCHy as a whole.

The starting question was whether a second economy could simply reproduce the SNoG.

A static technological model suggests that it might.

A strategic model gives a different answer.

The first SNoG changes the international system.

The second economy therefore does not confront the environment in which the first SNoG originally emerged.

Moreover, replication changes incumbent incentives.

Ordinary knock-out options transform the incumbent into a moving target.

Nuclear knock-out options introduce a catastrophic-risk term whose expected marginal cost can rise with strategic similarity.

The rival therefore solves

$$\max_{\mu} [V_B(\mu) - C_B(\mu) - R_B^K(\mu) - R_B^N(\mu)].$$

When

$$C'_B(\mu) + (R_B^K)'(\mu) + (R_B^N)'(\mu) > V'_B(\mu),$$

the marginal value of replication becomes negative.

Hence

$$\boxed{\mu_B^* < 1.}$$

Backward induction removes exact replication from the relevant subgame-perfect equilibrium path.

The First-Mover Non-Isomorphism Principle strengthens the result: even literal institutional identity does not recreate the original strategic object because

$$(\mathcal{S}_A, \Gamma_0) \neq (\mathcal{S}_A, \Gamma_1).$$

The deepest conclusion is therefore not merely that “there can be only one SNoG.”

The more precise statement is:

$$\boxed{\text{There can be only one original SNoG strategic position.}}$$

Other economies can learn from it.

They can imitate components.

They can cooperate with it.

They can construct related systems.

They can negotiate interoperability.

But once the original SNoG exists, the international state has changed.

Consequently, the strategic optimization problem faced by every subsequent economy is different.

The equilibrium logic of the international system therefore points toward

imitation $\rightarrow$ strategic interaction $\rightarrow$ endogenous counter-response $\rightarrow$ optimal differentiation.

The international-relations theory of the Standard Nuclear oliGARCHy is therefore ultimately a theory of endogenous uniqueness, deterrence, first-mover non-isomorphism, institutional evolution, strategic differentiation, and peaceful coexistence under catastrophic risk.

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Glossary

Anti-Replication Theorem

The result that exact SNoG replication is not optimal when the marginal strategic cost of convergence eventually exceeds its marginal benefit.

Catastrophic-Risk State

An extreme state associated with very large losses. In this paper it is modeled abstractly rather than operationally.

Complete Strategic Object

The pair $\tilde{\mathcal{S}} = (\mathcal{S}, \Gamma)$ consisting of domestic architecture and international environment.

Differentiation

The deliberate choice of a strategic architecture that remains distinct from the incumbent system.

Diplomatic Risk Transformation

The use of diplomacy to change the mapping from strategic similarity into perceived or objective strategic risk.

First-Mover Non-Isomorphism Principle

The result that the first SNoG changes the international state, preventing a subsequent institutional copy from occupying the same complete strategic state.

International State

The geopolitical, diplomatic, alliance, financial, institutional, and strategic environment represented by Γ .

Knock-Out Option

An inherited SNoG strategic option that changes the continuation value of a rival strategy after a relevant strategic condition is reached.

Moving Target

A system whose architecture changes in response to an attempt to copy it.

Nuclear Knock-Out Option

The previously derived SNoG nuclear deterrence option, represented in this paper only through its strategic option value and catastrophic-risk effects.

Replication Index

The variable $\mu \in [0, 1]$ measuring similarity between the incumbent SNoG and a prospective replicator.

Replication Paradox

The phenomenon in which increasing success at replication raises the countervailing strategic costs of further replication.

Security Dilemma

A condition in which actions intended as defensive by one state increase the perceived insecurity of another.

SNoG

The Standard Nuclear oliGARCHy.

Strategic Differentiation Equilibrium

An equilibrium in which rational economies select distinct architectures rather than converging exactly on the incumbent SNoG.

Strategic Non-Replicability

The inability to reproduce the complete strategic object as an equilibrium outcome even when observable institutions may be technologically reproducible.

Strategic Reflexivity

The feedback process through which replication changes incumbent behavior, which changes the environment, which changes the replicator's optimal policy.

Technological Replication

Reproduction of observable institutional, economic, political, financial, or technical components without necessarily reproducing the original strategic environment.

The End